

AI Validation Log

Project: AI-Assisted Insights for Indian Crop Production (1997–2015)

Dataset: Indian Crop Production Dataset

Objective: Evaluate AI-generated analytical insights against the actual dashboard and dataset before including them in the executive insight deck.

Part A – AI-Generated Insight Prompts

Prompt 1

Based on the dataset containing crop production across Indian states, identify the top crops contributing to overall agricultural production and explain their significance.

Evaluation: ✓ Accurate

Reason: The dashboard confirms that a small number of crops dominate production, with Coconut having the highest production among the recorded crops.

Prompt 2

Analyze whether agricultural production has increased or decreased over the years and identify any long-term production trends.

Evaluation: ✓ Accurate

Reason: The line chart clearly shows production trends across multiple years, making this insight directly supported by the data.

Prompt 3

Identify which agricultural season contributes the largest share of total production and explain why.

Evaluation: ~Partially Accurate

Reason: The dashboard shows production by season, but explaining *why* one season performs better would require additional climate and rainfall data that are not included in the dataset.

Prompt 4

Determine which Indian states contribute the highest agricultural production and discuss possible regional differences.

Evaluation: ~ Partially Accurate

Reason: State-level production can be measured from the dataset, but explaining regional differences would require information such as soil quality, irrigation facilities, and climate conditions.

Prompt 5

Recommend policy actions that could improve agricultural productivity based on the production data.

Evaluation: ~Partially Accurate

Reason: The dataset supports identifying production patterns, but policy recommendations should also consider economic, environmental, and social factors beyond production statistics.

Part B – AI Claims Edited During Report Writing

Example 1

AI Draft

Coconut is the most important crop in India.

Edited Version

Coconut records the highest production in this dataset; however, this reflects the available data and should not be interpreted as the overall economic importance of the crop across India.

Reason for Edit

The original statement overgeneralized beyond what the dataset supports.

Example 2

AI Draft

Higher production means higher agricultural efficiency.

Edited Version

Higher production indicates greater output but does not necessarily represent higher efficiency, as efficiency also depends on cultivated area, yield, resource utilization, and farming practices.

Reason for Edit

Production and efficiency are different performance measures.

Example 3

AI Draft

The dashboard proves that production increased every year.

Edited Version

The dashboard shows an overall long-term production trend, but yearly fluctuations are present and should be considered before concluding.

Reason for Edit

The original statement ignored year-to-year variability and overstated the evidence.

Validation Summary

Throughout this project, AI was used to generate analytical questions and draft initial insights. Every AI-generated statement was verified against the dashboard and the underlying dataset before being included in the final presentation. Where AI made assumptions beyond the available evidence, the statements were revised to ensure they accurately reflected the data and avoided unsupported conclusions.